

# Mathematical Reasoning

Centre sample · 5 questions with worked answers

This sample contains 5 questions from the Mathematical Reasoning section of an Aris practice paper, with full worked answers at the back. In the NSW Selective Test, Mathematical Reasoning has 35 questions in 40 minutes. Allow about 6 minutes for this sample.

**Instructions:** Choose the best answer for each question and circle the letter. Calculators are not allowed. Working can be shown in the space around each question.

## Question 1

Each symbol below represents a different number.

$$\begin{aligned}4 \times \star &= 20 \\ \bigcirc &= \star + 10 \\ \triangle &= \bigcirc - 6\end{aligned}$$

What is the value of  $\triangle$ ?

A. 5

B. 8

C. 9

D. 10

E. 15

## Question 2

Emma had \$20. Emma gave some money to Liam. Liam then gave half of what he was given to Oliver. Oliver gave two thirds of what he was given to Jason. Jason received \$5.

How much did Emma give to Liam?

A. \$2.50

B. \$5

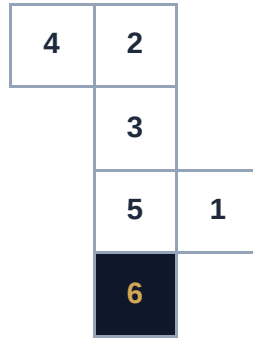
C. \$7.50

D. \$15

E. \$15.50

### Question 3

The net below folds into a cube. Which number is on the face opposite 6?



A. 1

B. 2

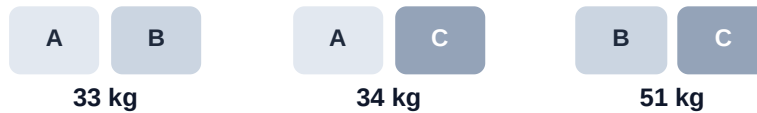
C. 3

D. 4

E. 5

### Question 4

Three different boxes are weighed in pairs. The combined weights of the three pairs are **33 kg**, **34 kg** and **51 kg**, as shown.



When the three boxes are weighed together, what is their combined weight?

A. 59 kg

B. 67 kg

C. 84 kg

D. 85 kg

E. 118 kg

### Question 5

In a magic square, each row, column and diagonal adds up to the same total. Some numbers are missing.

|    |    |    |
|----|----|----|
| 10 | 30 | 26 |
| 38 |    |    |
|    | ?  |    |

What is the missing number at ?

A. 13

B. 14

C. 15

D. 17

E. 18

## Worked answers (for tutors)

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### Question 1

$4 \times \star = 20$ ,  $\bigcirc = \star + 10$ ,  $\triangle = \bigcirc - 6$ . What is  $\triangle$ ?

$$\begin{aligned}\star &= 20 \div 4 = 5 \\ \bigcirc &= 5 + 10 = 15 \\ \triangle &= 15 - 6 = 9\end{aligned}$$

Check:  $4 \times 5 = 20$ ;  $5 + 10 = 15$ ;  $15 - 6 = 9$ . All three equations hold.

**Answer: C (9)**

### Question 2

Emma gave some money to Liam. Liam gave half to Oliver. Oliver gave two thirds of what he was given to Jason. Jason received \$5. How much did Emma give to Liam?

Let  $x$  = the amount Emma gave Liam.  
Liam gives Oliver:  $x \div 2 = x/2$   
Oliver gives Jason:  $(2/3) \times (x/2) = x/3$   
Jason received \$5, so  $x/3 = 5$   
 $x = 3 \times 5 = 15$

Check: \$15  $\rightarrow$  half is \$7.50  $\rightarrow$  two thirds of \$7.50 is \$5. The chain holds.

**Answer: D (\$15)**

### Question 3

The net folds into a cube. Which number is on the face opposite 6?

Start with the four faces in the straight vertical strip: 2, 3, 5 and 6. When the net folds, this strip wraps around the cube like a belt.

Strip around the cube: 2, 3, 5, 6, then back to 2  
Faces two places apart are opposite:  
2 is opposite 5  
3 is opposite 6

The remaining faces, 4 and 1, close the two ends and are opposite each other.

Therefore, the face opposite 6 is 3.

**Answer: C (3)**

**Teaching tip:** Look for a straight strip of four faces. It wraps around the cube like a belt, so faces two positions apart in the strip are opposite.

#### Question 4

$A + B = 33$  kg,  $A + C = 34$  kg,  $B + C = 51$  kg. What do all three boxes weigh together?

Add the three pairings:

$$(A + B) + (A + C) + (B + C) = 33 + 34 + 51$$

$$2A + 2B + 2C = 118$$

$$A + B + C = 118 \div 2 = 59 \text{ kg}$$

Individual weights (not needed, but a good check):

$$A = 59 - 51 = 8 \text{ kg}$$

$$B = 59 - 34 = 25 \text{ kg}$$

$$C = 59 - 33 = 26 \text{ kg}$$

Check:  $8 + 25 = 33$ ;  $8 + 26 = 34$ ;  $25 + 26 = 51$ . All three pairings hold.

**Answer: A (59 kg)**

**Teaching tip:** Each box is weighed exactly twice across the three pairings, so the three pair totals add to double the combined weight. Students who see this skip solving for A, B and C entirely.

#### Question 5

Every row, column and diagonal adds to the same total. What is the missing number at ?

Row 1 is complete:  $10 + 30 + 26 = 66$  (the magic total)

Column 1:  $10 + 38 + \_ = 66 \rightarrow 18$

Diagonal:  $26 + \_ + 18 = 66 \rightarrow \text{centre} = 22$

Row 2:  $38 + 22 + \_ = 66 \rightarrow 6$

Column 2:  $30 + 22 + ? = 66 \rightarrow ? = 14$

The completed square:

|    |    |    |
|----|----|----|
| 10 | 30 | 26 |
| 38 | 22 | 6  |
| 18 | 14 | 34 |

Check: every row, column and diagonal adds to 66.

**Answer: B (14)**

**Answer key: Q1 C · Q2 D · Q3 C · Q4 A · Q5 B**

**About Aris practice papers:** Full papers cover Thinking Skills, Reading, Mathematical Reasoning and Writing in exam format, each with a complete answer section. Email [contact@aris-academy.com](mailto:contact@aris-academy.com) for the other free samples and licence details.